

## KEMI 1 FYR, Finland

WMO: 028630

Lat: 65.3853N

Lon: 24.0961E

Elev: 19

StdP: 101.10

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | -20.7      | -17.9 | -23.6                         | 0.4 | -20.6 | -20.3 | 0.6 | -17.8 | 18.2                  | -2.0 | 17.1 | -2.6 | 4.4                      | 10   | 0.931 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 3.2                    | 21.1            | 17.6 | 19.5 | 16.4 | 18.2 | 15.7 | 18.2                | 20.3 | 17.0 | 18.9 | 16.1 | 17.9 | 5.6                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.3                            | 12.4 | 19.6 | 16.2 | 11.5 | 18.2 | 15.2 | 10.8 | 17.3 | 51.7          | 20.5 | 47.9 | 19.0 | 45.3 | 18.0 | 21.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 17.2          | 15.7 | 14.3 | DB                         | -22.4 | 22.6               | 4.9 | 1.9                                                | -25.9 | 24.0       | -28.7 | 25.1       | -31.5 | 26.2       | -35.0 | 27.6 | (4) |
| (5)               |      |      | WB                         | -22.4 | 18.8               | 4.7 | 1.6                                                | -25.8 | 20.0       | -28.5 | 21.0       | -31.1 | 21.9       | -34.5 | 23.1 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 3.9   | -5.8  | -7.3  | -4.1  | 0.4   | 5.9   | 10.9  | 15.4  | 14.8  | 11.0  | 5.5   | 1.3   | -1.5 | (6)  |
| (7)  |                                                                                  | DBStd       | 8.45  | 5.32  | 6.27  | 4.26  | 2.58  | 3.43  | 2.82  | 2.66  | 2.37  | 2.37  | 3.01  | 3.62  | 4.58 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2636  | 489   | 484   | 438   | 288   | 136   | 22    | 1     | 1     | 17    | 142   | 262   | 357  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5264  | 747   | 718   | 696   | 538   | 386   | 224   | 98    | 112   | 220   | 399   | 512   | 615  | (9)  |
| (10) |                                                                                  | CDD10.0     | 422   | 0     | 0     | 0     | 0     | 8     | 49    | 168   | 149   | 47    | 1     | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 11    | 0     | 0     | 0     | 0     | 0     | 1     | 8     | 2     | 0     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 13    | 0     | 0     | 0     | 0     | 0     | 3     | 10    | 0     | 0     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.4   | 8.1   | 7.3   | 7.2   | 6.3   | 6.7   | 6.6   | 6.1   | 6.8   | 7.8   | 8.4   | 8.8   | 9.1  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 601   | 48    | 41    | 34    | 32    | 42    | 50    | 66    | 63    | 57    | 56    | 61    | 53   | (15) |
| (16) |                                                                                  | PrecMax     | 823   | 90    | 88    | 60    | 80    | 87    | 97    | 129   | 144   | 118   | 103   | 111   | 120  | (16) |
| (17) |                                                                                  | PrecMin     | 452   | 13    | 5     | 10    | 8     | 10    | 8     | 10    | 13    | 15    | 4     | 17    | 17   | (17) |
| (18) |                                                                                  | PrecStd     | 114   | 22    | 22    | 15    | 20    | 17    | 25    | 29    | 36    | 31    | 25    | 27    | 26   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 3.2   | 3.6   | 3.9   | 9.1   | 19.1  | 21.0  | 24.8  | 21.4  | 16.2  | 11.9  | 8.9   | 5.3  | (19) |
| (20) |                                                                                  |             | MCWB  | 2.2   | 1.9   | 1.8   | 4.2   | 12.5  | 15.1  | 19.4  | 18.5  | 14.1  | 10.1  | 7.3   | 3.9  | (20) |
| (21) |                                                                                  | 2%          | DB    | 2.2   | 2.4   | 2.7   | 6.3   | 15.4  | 17.3  | 21.9  | 19.9  | 15.2  | 10.9  | 7.8   | 4.6  | (21) |
| (22) |                                                                                  |             | MCWB  | 1.6   | 1.1   | 1.1   | 3.1   | 10.7  | 13.1  | 18.3  | 16.9  | 13.4  | 9.5   | 6.7   | 3.4  | (22) |
| (23) |                                                                                  | 5%          | DB    | 1.5   | 1.2   | 1.7   | 4.4   | 12.5  | 15.7  | 20.5  | 18.8  | 14.6  | 10.1  | 6.8   | 4.0  | (23) |
| (24) |                                                                                  |             | MCWB  | 0.9   | 0.4   | 0.5   | 2.1   | 8.7   | 12.5  | 17.3  | 16.4  | 12.9  | 8.8   | 5.8   | 3.0  | (24) |
| (25) |                                                                                  | 10%         | DB    | 0.6   | 0.2   | 1.0   | 3.2   | 10.3  | 14.5  | 19.0  | 17.9  | 14.1  | 9.4   | 5.7   | 3.4  | (25) |
| (26) |                                                                                  |             | MCWB  | 0.1   | -0.3  | 0.1   | 1.4   | 7.3   | 12.0  | 16.4  | 15.7  | 12.5  | 8.0   | 4.6   | 2.5  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 2.5   | 2.3   | 2.2   | 5.5   | 13.6  | 16.0  | 20.3  | 20.0  | 14.7  | 10.7  | 7.8   | 4.6  | (27) |
| (28) |                                                                                  |             | MCDWB | 2.9   | 3.4   | 3.3   | 7.5   | 17.2  | 19.2  | 22.6  | 21.2  | 15.8  | 11.3  | 8.3   | 4.9  | (28) |
| (29) |                                                                                  | 2%          | WB    | 1.8   | 1.4   | 1.4   | 3.6   | 11.1  | 14.5  | 19.2  | 17.5  | 14.0  | 10.0  | 7.0   | 3.7  | (29) |
| (30) |                                                                                  |             | MCDWB | 2.1   | 2.2   | 2.3   | 5.9   | 15.3  | 16.5  | 21.7  | 19.1  | 15.0  | 10.8  | 7.7   | 4.2  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.1   | 0.5   | 0.8   | 2.5   | 9.1   | 13.3  | 17.8  | 16.8  | 13.3  | 9.1   | 5.9   | 3.2  | (31) |
| (32) |                                                                                  |             | MCDWB | 1.5   | 1.1   | 1.5   | 3.8   | 11.5  | 15.1  | 20.1  | 18.4  | 14.4  | 10.0  | 6.6   | 3.9  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.2   | -0.1  | 0.2   | 1.8   | 7.8   | 12.3  | 16.6  | 16.0  | 12.8  | 8.2   | 4.9   | 2.5  | (33) |
| (34) |                                                                                  |             | MCDWB | 0.6   | 0.4   | 0.9   | 2.9   | 10.2  | 14.0  | 18.7  | 17.6  | 13.9  | 9.2   | 5.6   | 3.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 4.2   | 4.6   | 4.7   | 3.9   | 4.7   | 3.7   | 3.2   | 2.6   | 2.4   | 2.6   | 3.1   | 3.4  | (35) |
| (36) |                                                                                  |             | MCDBR | 3.2   | 3.6   | 3.7   | 5.4   | 8.8   | 5.8   | 4.7   | 3.3   | 2.5   | 2.5   | 3.0   | 3.0  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.2   | 3.3   | 3.1   | 3.5   | 4.7   | 3.2   | 3.0   | 2.7   | 2.4   | 2.4   | 3.2   | 2.9  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 3.2   | 3.7   | 3.4   | 4.8   | 7.9   | 5.0   | 3.9   | 2.9   | 2.3   | 2.5   | 2.8   | 3.0  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.2   | 3.5   | 3.0   | 3.4   | 4.8   | 3.5   | 3.0   | 2.6   | 2.2   | 2.5   | 3.0   | 2.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.207 | 0.271 | 0.300 | 0.336 | 0.343 | 0.348 | 0.366 | 0.352 | 0.328 | 0.305 | 0.343 | 0.353 | (40) |      |
| (41) |                                                                                  | taud        | 1.943 | 2.075 | 2.116 | 2.143 | 2.400 | 2.499 | 2.468 | 2.498 | 2.538 | 2.353 | 2.559 | 2.595 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 271   | 590   | 758   | 821   | 862   | 867   | 835   | 814   | 756   | 572   | 255   | 93    | (42) |      |
| (43) |                                                                                  | Edh at Noon | 21    | 60    | 92    | 115   | 99    | 92    | 93    | 82    | 64    | 47    | 20    | 10    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.09  | 0.60  | 2.12  | 3.89  | 5.06  | 5.70  | 5.25  | 3.85  | 2.07  | 0.75  | 0.14  | 0.02  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.01  | 0.12  | 0.21  | 0.27  | 0.45  | 0.43  | 0.45  | 0.35  | 0.23  | 0.13  | 0.02  | 0.00  | (45) |      |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (42 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page